# Jason Wakim Richards

+44 77 08 348 077 | jason2richards@gmail.com
github.com/richarJ2002 | linkedin.com/in/jason-richards-jwr2002/ | richarj2002.github.io/personal_website/

Guidance, navigation, and controls engineer with a First-Class MEng in Aerospace Engineering and hands-on experience in aircraft dynamics, embedded flight software, rigid-body simulation, digital signal processing, UAV flight testing, and integrated GNC test campaigns. I develop C++ and Python autonomy software for physical UAVs and evaluate it through simulation, laboratory, hardware, and flight testing. At Airbus Defence & Space, I developed C/C++ GNC simulation software and led LiDAR test activities for an autonomous rover. As ADCS Lead for the JamSail 3U CubeSat, I developed embedded flight software, an attitude-dynamics and control simulation, an EKF, and digital sensor filters. My background combines longitudinal and lateral aircraft-dynamics analysis against flight-test data with test planning, cross-disciplinary integration, performance analysis, and verification and validation.

## Work Experience

### Doctoral Researcher – Autonomous UAV Systems

November 2025 – Present

**SnT, University of Luxembourg** | Kirchberg, Luxembourg

- Develop high-quality C++ and Python autonomy software for UAV localization, Semantic Relational SLAM, and situational awareness, linking camera/IMU perception and state estimation with planning and control.
- Integrate, tune, and deploy autonomous navigation on physical UAVs using ROS 2, PX4, embedded Linux, Raspberry Pi, and OpenVINS visual–inertial odometry, directly supporting controlled flight trials.
- Build reproducible Docker-based development and deployment environments and camera–IMU characterization and calibration workflows with Kalibr, supported by version-controlled software and technical documentation.
- Formulate test plans and conduct simulation, laboratory, hardware, and flight testing to assess integrated performance and robustness across sensing, estimation, mapping, planning, and control.
- Collaborate closely across software, controls, perception, and vehicle-integration activities, taking ownership of implementation, interface troubleshooting, test analysis, and technical documentation.

### AOCS/GNC Engineer Intern

July 2023 – August 2024

**Airbus Defence & Space** | Stevenage, UK

- Served as lead test engineer for LiDAR activities during Integrated BreadBoarding (iBB) 3 & 4, formulating procedures and supporting end-to-end verification of the Sample Fetch Rover concept at Technology Readiness Level 6.
- Designed and built an integrated hardware/software GNC test bench, connecting LiDAR to rover breadboards through ROS over a local wireless network to capture data, characterize sensor performance, and validate navigation models.
- Developed C/C++ software on Linux for an offline GNC simulation, coupling a natural-scene generator to produce synthetic camera imagery and LiDAR scans for rapid algorithm development and performance analysis.
- Investigated LiDAR and radar simulation, terrain generation, and Vector Field Histogram planning for lunar-rover navigation, translating prototype concepts into the existing simulation environment.
- Analyzed test-campaign data and converted results into technically grounded integration actions while collaborating across GNC, simulation, software, and test disciplines in a fast-paced Agile team.

## Projects & Leadership

### JamSail 3U CubeSat – AOCS/ADCS Lead

September 2024 – August 2025

**NottSpace, University of Nottingham** | Nottingham, UK

- Led ADCS design and implementation for a NottSpace 3U CubeSat, translating the concept of operations into subsystem and algorithm requirements, control modes, architecture, interfaces, and a verification and validation plan.
- Developed embedded C flight software and a C/C++/CMake Linux simulation of rigid-body attitude dynamics and control to design and verify GNC algorithms before deployment to flight hardware.
- Designed and implemented an EKF that fuses sun-sensor, magnetometer, and IMU measurements, together with TRIAD attitude determination and embedded control-state logic.
- Performed FFT-based sensor-signal analysis, selected filter cut-off frequencies, implemented first-order digital low-pass filters, and verified performance through simulation and hardware testing.

### IDET-Sat – Systems Engineer
September 2022 – July 2023

**Nottingham Space Society** | Nottingham, UK

- Defined system requirements, architecture, subsystem interfaces, Project Breakdown Tree, and Work Breakdown Structure within a student-led, 12-person team developing a 3U CubeSat.
- Developed software on a Raspberry Pi running Linux and integrated a microcontroller, IMU, and cameras over I2C before the concept progressed to the competition build phase.
- Contributed across embedded software, hardware and sensor integration, and system simulation while coordinating interfaces and technical development within the student team.
- Achieved a top-four finish among 12 UK teams as first-time entrants to the UKSEDS Satellite Design Competition.
- Led cross-team meetings, communicated design changes, and presented the integrated concept to industry experts, balancing technical interfaces, build constraints, and competition deadlines.

### UKSEDS Volunteer
Sept. 2024 – Sept. 2026

**UK Students for the Exploration and Development of Space** | Remote & UK field events

- Served as an Olympus Rover Trials (ORT) volunteer in the first year, helping organise a UK national rover competition with Airbus Defence & Space and RAL Space.
- Progressed in the second year to Competitions Management Co-Lead, overseeing four national competitions: ORT, SDC, ISAM, and NRC.
- Coordinated volunteers, schedules, technical milestones, participant support, and partner engagement with Airbus Defence & Space (SDC), the UK Space Agency (ISAM), Midlands Rocketry Club, and Skyrora (NRC).
- Produced clear participant documentation and supported venue and field-event logistics, helping multidisciplinary volunteer teams deliver competitions to fixed schedules.

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## Education & Training

### MEng Aerospace Engineering – First-Class Honours
September 2020 – August 2025

**University of Nottingham** | Nottingham, UK

- Graduated with First-Class Honours, achieving 99% in Dynamics & Flight Mechanics and 91% in Avionics Systems, with specialization in Aerospace Control Systems.
- Developed MATLAB tools to analyze longitudinal and lateral aircraft dynamics, comparing modeled motion and dynamic modes with Cranfield University flight-test data.
- Applied C/C++, Python, MATLAB, Simulink, and Linux to control design, flight-dynamics modeling, stability and performance analysis, estimation, and simulation.

### IEEE RAS Summer School on Multi-Robot Systems
July 2026

**Czech Technical University in Prague** | Prague, Czechia

- Completed the 2026 IEEE RAS Summer School on Multi-Robot Systems, covering cooperative aerial robotics and active perception; presented the SaCoGraph collaborative-UAV research concept.
- Applied multi-robot autonomy concepts to collaborative sensing, semantic mapping, and resource-aware mission planning.

### Space Resources & Robotics Summer School
May 2026

**AGH University of KrakÃşw** | Krakow, Poland

- Completed intensive training in space resources, planetary robotics, and robotic manipulation for exploration applications.
- Developed a broader systems perspective on robotic operations in challenging off-world environments.

### Space Systems Engineering Training Course
May 2025

**ESA Academy** | Transinne, Belgium

- Completed intensive ESA Academy training across the space-system lifecycle, from mission objectives and concepts of operations through architecture, integration, verification, and operations.
- Applied requirements engineering and concurrent-design methods to translate user needs into traceable, testable requirements and manage interfaces between technical disciplines.
- Practised system options and trade studies, budgets and margins, verification and validation planning, and the relationship between engineering decisions, schedule, cost, and risk.
- Completed multidisciplinary group exercises and communicated design rationale and systems-engineering deliverables, strengthening collaborative problem-solving across complex spacecraft projects.